Name:	CT :	Index no:

PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2008

H1 BIOLOGY
PAPER 2

8875/02
10 Sept 2008
2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

This document consists of 2 sections:	**For Examiner's Use**	
	Section A	
Section A (50 marks)	**1**	**/ 10**
Answer **all** questions.	**2**	**/ 10**
Answers are to be written in the spaces provided on the question paper.	**3**	**/ 10**
	4	**/ 10**
	5	**/ 10**
	Sub-total	**/ 50**
Section B (20 marks)	**Section B**	
Answer **one** question.	**6 or 7**	**/ 20**
	TOTAL	

Do not open this booklet until you are told to do so.

This document consists of 13 printed pages (inclusive of this page)

Section A: Structured Questions

Question 1 [10 marks]

(a) Fig 1.1 below shows the results of an investigation into the effect of a competitive inhibitor on an enzyme-controlled reaction over a large range of substrate concentrations.

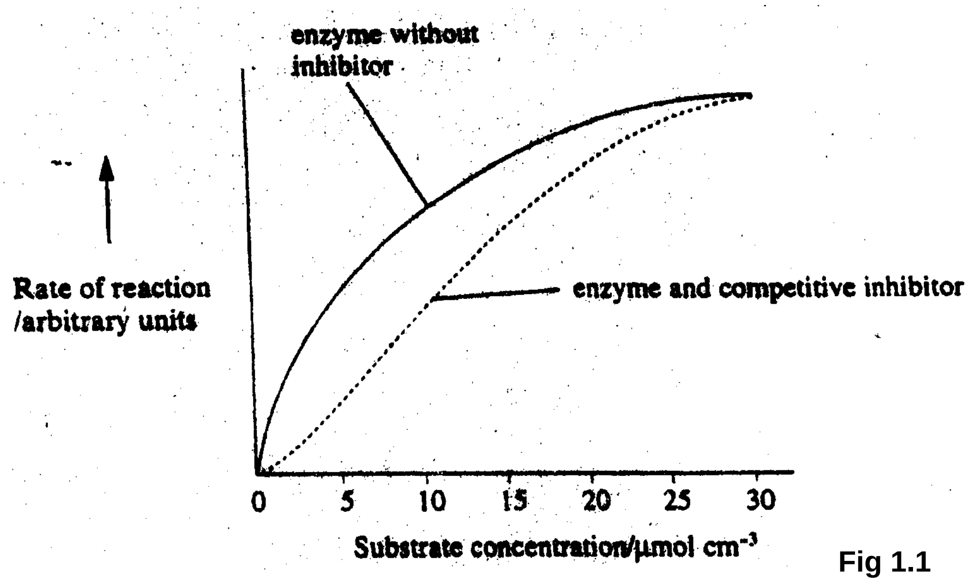

Fig 1.1

(i) Explain the difference in the rates of reaction between the two experimental conditions at the substrate concentration of 10 umol cm^{-3}. [3]

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(ii) Explain why the rates of reaction at the two experimental conditions are similar at the substrate concentration of 30 umol cm^{-3}. [2]

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Alcohol dehydrogenase is an enzyme found in the liver. It normally breaks down ethanol (C_2H_5OH) into less harmful products. About 50 deaths each year occur following ingestion of a compound called ethylene glycol ($C_2H_4(OH)_2$) which is found in antifreeze. This compound is not lethal, but it is broken down by alcohol dehydrogenase into highly toxic oxalic acid. Administering a large dose of ethanol as quickly as possible can treat ethylene glycol poisoning.

(b) Use your knowledge of enzyme activity to explain how this treatment may counteract the effects of ethylene glycol poisoning. [2]

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(c) Explain what causes a particular enzyme molecule to always fold into the same shape. [3]

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Question 2 [12 marks]

(a) During embryo development, the cells start to differentiate after the blastula stage.

Describe a DNA modification which enables the cells to differentiate. [2]

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(b) Protein synthesis is necessary for the survival of cells where important proteins are synthesised. Protein synthesis involves both the process of transcription and translation. Any disruption to these processes may cause the cell to be dysfunctional.

(i) Describe the importance of the DNA-binding domain and protein-binding domain of transcription factors. [2]

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(ii) A retrotransposon (a 'jumping gene' containing non-coding DNA) was accidentally inserted at the TATA region in the promoter of a gene. Explain how this would affect transcription of that gene. [2]

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(iii) State precisely where in the cell translation occurs. [1]

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(iv) Describe how the structure of the ribosome is adapted for translation. [3]

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(v) A mutation was found in a gene that codes for the capping enzyme complex. Explain how this mutation would affect translation. [2]

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Question 3 [10 marks]

(a) Coat colour in cats is determined by a sex-linked gene with two alleles, black and orange. When black cats are mated with orange cats, the female offspring are always tortoiseshell, their coats show black and orange patches of various sizes, while the male offspring have the same coat colour as their mothers.

Using the appropriate symbols, draw genetic diagrams in the space below to account for this cross. [4]

(b) A cross was made between two fruit flies. One parent had an unknown genotype since it had the wild phenotype (normal [long] wings and normal body colour). The other parent had vestigial [underdeveloped] wings and ebony body colour, so it was known to be homozygous recessive for both characteristics.

The resulting offspring were as follows:

Normal wings and normal body colour	45
Normal wings and ebony body colour	48
Vestigial wings and normal body colour	44
Vestigial wings and ebony body colour	51

Using the following symbols:

N normal wings	n vestigial wings
E normal body	e ebony body

(i) Draw a genetic diagram in the space below to show the cross described. [4]

(ii) Vestigial wing and ebony body are caused by artificially induced mutations in the laboratory bred fruit flies. Suggest why such mutations are unlikely to be found in natural populations. [2]

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Question 4 [8 marks]

Fig. 4.1 below shows the main stages in the polymerase chain reaction (PCR).

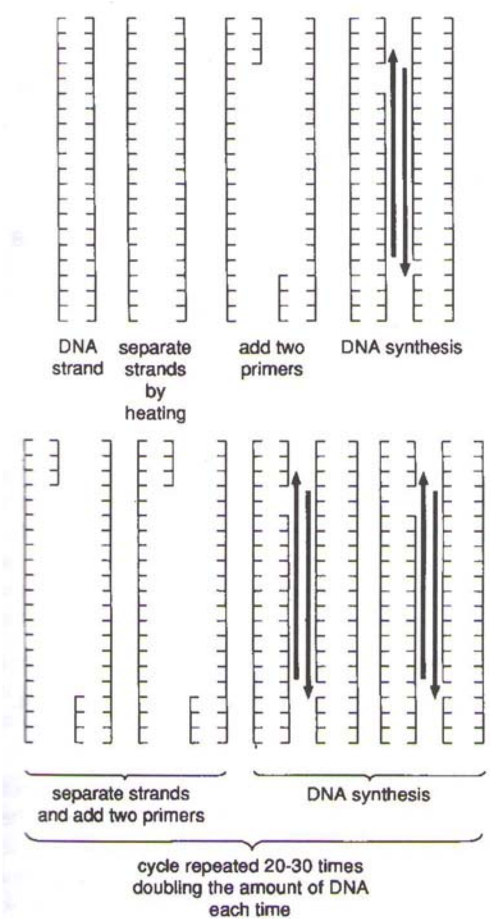

Fig 4.1

(a) Suggest why the brief heat treatment separates two strands of DNA double helix. [2]

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(b) Describe the structure and function of a primer in PCR. [2]

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(c) Explain the advantages of using Taq DNA polymerase instead of DNA polymerase. [2]

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(d) The copying error rate for PCR is much higher as compared to DNA replication *in vivo*. Suggest why this is so. [2]

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Question 5 [10 marks]

(a) Fig 5.1 below is an electron micrograph of a mesophyll cell from a leaf.

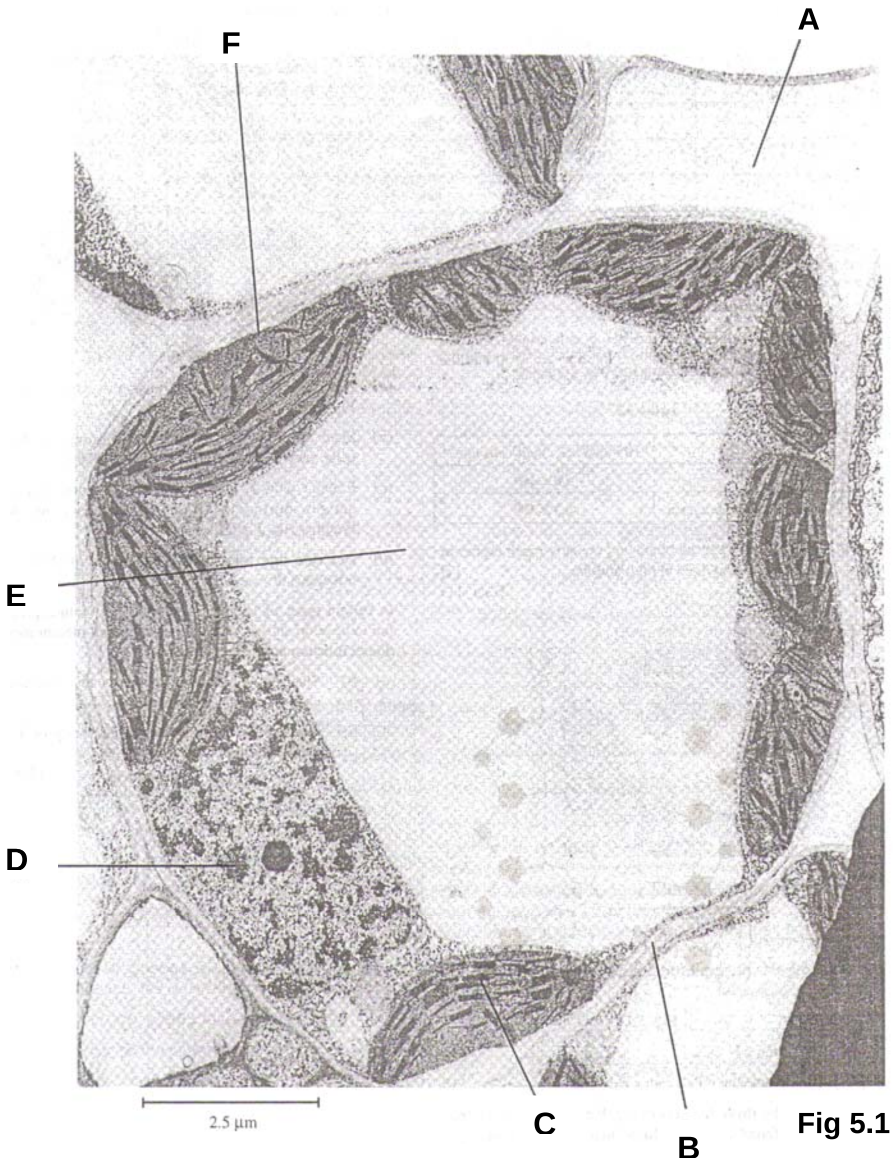

Fig 5.1

(i) Identify, by using the letters A to F, a part of the cell shown in Fig 5.1 where the following substances are located. [2]

chlorophyll ………………………………………………….

cellulose ………………………………………………….

DNA ………………………………………………….

phospholipids ………………………………………………….

(ii) State precisely where the light independent reactions (Calvin cycle) occur in plant cells. [1]

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The pathway of carbon during the reactions of photosynthesis was discovered using the unicellular green algae Chlorella. The Chlorella culture was supplied with $^{14}CO_2$ and illuminated with varying length of time. Some algae were removed at intervals and immediately killed by boiling in alcohol. This provided an extract which was examined using chromatographic techniques and any radioactive compounds present showed up on X-ray film as blackened areas.

After 15 seconds of illumination, the extract produced three prominent blackened areas which were due to the presence of three compounds:

i. glycerate phosphate (GP/ PGA)
ii. ribulose biphosphate (RuBP)
iii. triose phosphate (TP/ PGAL)

(b) In a second series of experiments the algae were illuminated in the presence of $^{14}CO_2$ for 15 seconds. Illumination of the algae continues for a further 1 minute after all $^{14}CO_2$ present had been removed. Some algae were then removed, killed and the extract examined. A distinct blackened area was identified on the X-ray film.

Name the compound that was found in this blackened area. Explain your answer. [2]

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(c) In another series of experiments, the algae were illuminated in the presence of $^{14}CO_2$ for 15 seconds and then placed in the dark for 1 min in the presence of $^{14}CO_2$. Some algae were removed, killed and the extract examined. A distinct blackened area was identified on the X-ray film.

Name the compound that was found in this blackened area. Explain your answer. [2]

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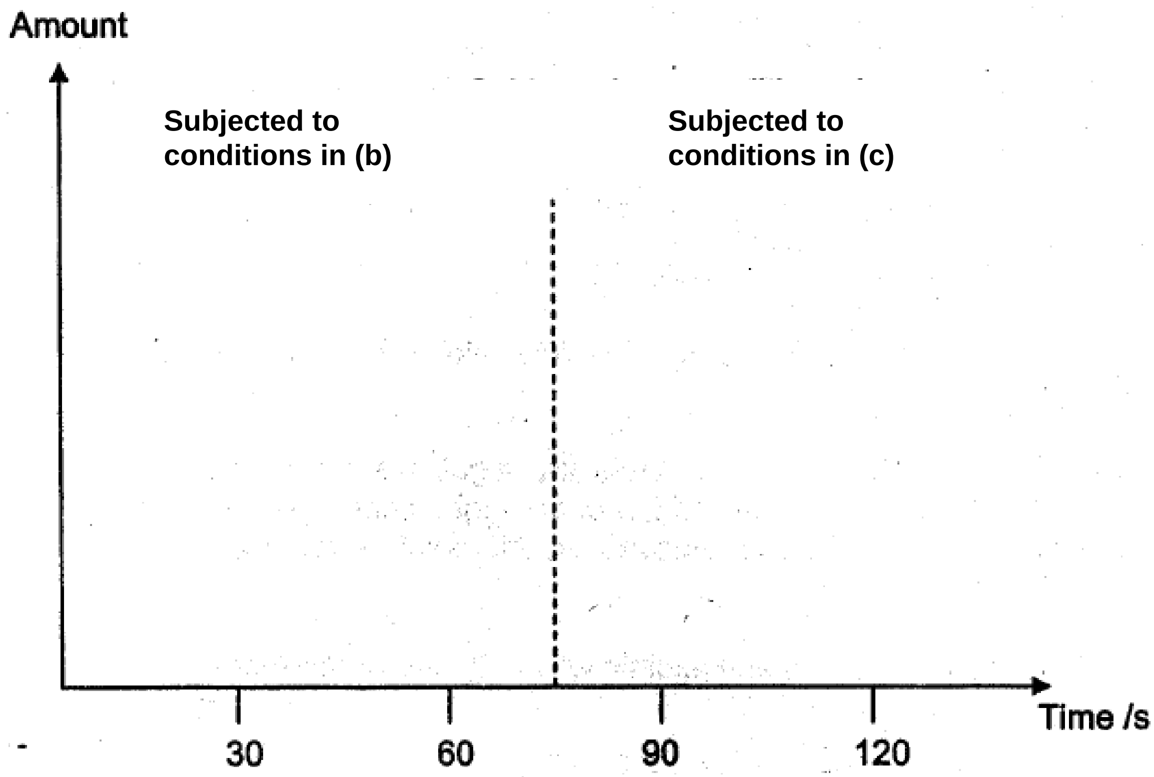
(d) On the graph below, show how the amounts of the labelled compounds named in (b) and (c) will change if the algae is *first* subjected to the conditions in (b) *then* to conditions in (c). [3]



Section B
Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate. Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

<u>Question 6</u>

(a) Describe, using examples, the roles of lipids in organisms. [10]

(b) Explain why lipids are better storage molecules than carbohydrates. [7]

(c) How do lipids differ structurally from carbohydrates? [3]

<u>Question 7</u>

(a) Describe the process of obtaining a named protein product using genetic engineering. [10]

(b) Describe the goals, benefits and ethical concerns of the Human Genome Project. [10]